

Exercise Sheet 12e - Solutions

Predicate Logic – Natural Deduction & Semantics

1. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow \forall \neg x < 0$.

$$\begin{array}{c}
 \frac{\overline{S_1}^1 \quad \frac{\overline{x < 0}^2}{\exists x. x < 0} [\exists I]}{\neg x < 0} [\neg E] \\
 \frac{\perp}{\neg x < 0}^2 [\neg I] \\
 \frac{\neg x < 0}{\forall x. \neg x < 0} [\forall I] \\
 \frac{\forall x. \neg x < 0}{(S_1) \rightarrow \forall x. \neg x < 0}^1 [\rightarrow I]
 \end{array}$$

2. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow \exists x. 0 < x$

$$\begin{array}{c}
 \frac{\overline{S_2}^2}{\exists y. 0 < y \vee y < 0} [\vee E] \quad \frac{\overline{0 < y \vee y < 0}^3 \quad \frac{\overline{0 < y}^4}{0 < y \rightarrow 0 < y}^4 [\rightarrow I] \quad \frac{\overline{S_1}^1 \quad \frac{\overline{y < 0}^5}{\exists x. x < 0} [\exists I]}{\perp} [\neg E]}{0 < y} [\perp E] \\
 \frac{0 < y}{y < 0 \rightarrow 0 < y}^5 [\rightarrow I] \\
 \frac{0 < y}{\exists x. 0 < x} [\exists I] \\
 \frac{\exists x. 0 < x}{(S_2) \rightarrow \exists x. 0 < x}^2 [\rightarrow I] \\
 \frac{(S_2) \rightarrow \exists x. 0 < x}{(S_1) \rightarrow (S_2) \rightarrow \exists x. 0 < x}^1 [\rightarrow I]
 \end{array}$$

3. For example, the following models are models of S_2 :

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid \text{True} \}, \emptyset \rangle \rangle$
- $M'_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \emptyset \rangle \rangle$
- $M''_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n > m \}, \emptyset \rangle \rangle$
- $M'''_1 = \langle \{0\}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid \text{True} \}, \emptyset \rangle \rangle$

4. For example, the following models are not models of S_2 , i.e., are models of $\neg S_2$:

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \emptyset \rangle \rangle$
- $M'_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle 0, 1 \rangle \}, \emptyset \rangle \rangle$